

As the state u_s depends on control f_s , then u_s can also be written as $u_s = u_s(f_s)$ and so from now we can use notation $J_s(f_s)$ instead of $J_s(u_s(f_s), f_s)$. In the above problem 3.1,3.2,3.3 we are going to find a optimal control $\bar{f}_s \in U_{ad} \subset L^2(\Omega)$ in a such a way that the corresponding solution of $\bar{u}_s$ together with $\bar{f}_s$ satisfies the minimization of the cost function, i.e.,

$$J_s(\bar{f}_s) := \min_{f_s \in U_{ad}} J_s(f_s),$$

and corresponding optimal control of classical Poisson equation with homogeneous boundary conditions is the following

$$\min J(u, f) = \frac{1}{2} \left(\|\nabla u\|_{L^2(\Omega)}^2 + \mu \|f\|_{L^2(\Omega)}^2 \right) \quad (3.4)$$

subject to (the *Poisson equation*)

$$\begin{cases} (-\Delta)u = f, & x \in \Omega, \\ u = 0, & x \in \partial\Omega, \end{cases} \quad (3.5)$$

and the control constraints

$$a \leq \|f\|_{L^2(\Omega)} \leq b. \quad (3.6)$$

Our main objective is to find the optimal control $\bar{f}$ such that

$$J(\bar{f}) = \min_{f \in U_{ad}} J(f).$$

Proposition 3.1 (see [3]). *Let $\mathcal{F}_s = \{f_s\}_{0 < s < 1} \subset H^{-s}(\Omega)$ be the sequence satisfying $\|f_s\|_{H^{-s}(\Omega)}$ uniformly bounded with respect to s and $f_s \rightharpoonup f$ weakly in $H^{-1}(\Omega)$ as $s \rightarrow 1^-$, then $u_s \rightarrow u$ strongly in $H_0^{1-\delta}(\Omega)$ for some $C > 0$ and $0 < \delta \leq 1$.*

Here, we will extend this proposition 2.2 to optimal control of fractional PDE in our main result.

The first proposition represents about the Poincaré inequality in fractional Sobolev space and the second refers to the minimizer of a function defined on a suitable space.

Proposition 3.2 (Poincaré inequality, [4]). *Let $s \in (0, 1)$, $\Omega \subset \mathbb{R}^N$ be an open and bounded set then we have*

$$\|u\|_2^2 \leq C(N, \Omega, s) [u]_{H^s(\Omega)}^2.$$

where some constant $C(N, \Omega, s)$ depending upon N, Ω and s .

Proposition 3.3. *Let the admissible control space U_{ad} be weakly closed, bounded subset of $L^2(\Omega)$ with $J_s : U_{ad} \rightarrow \mathbb{R}$ is weakly lower semi-continuous. Then J_s has minimizer in U_{ad} .*

The existence and uniqueness of optimal control via strict convex and lower semi-continuous is discussed in the following proposition.